

Question:

Calculate the quantity of heat required to (a) convert a liter of H₂O at 30°C to a liter of water at 60°C, and (b) heat a 1 kg block of aluminum from 30° to 60°C. Assume the specific heat of water and aluminum is, re-spectively, 1 cal/g°C and .215 cal/g°C.

Answer Choices:

- A. 28 Kcal, 6.3 Kcal
- B. 40 Kcal, 5 Kcal
- C. 30 Kcal, 8.58 Kcal
- D. 25 Kcal, 8 Kcal
- E. 29 Kcal, 7.77 Kcal
- F. 34 Kcal, 6.9 Kcal
- G. 30 Kcal, 6.45 Kcal
- H. 32 Kcal, 7 Kcal
- I. 35 Kcal, 7.5 Kcal
- J. 31 Kcal, 6.15 Kcal

Reasoning + Answer:

<think>

For (a), we need to calculate the heat to convert the temperature of 1 liter of water from 3 0 °C to 6 0 °C . Since the volume of water doesn 't change , we use the specific heat capacity of water which is 1 cal/g °C . First , convert 1 liter to grams (approximately 1 0 0 0 g), and use the formula : $q = m \cdot c \cdot \Delta T$

For (b), we do the same for the aluminum block with its specific heat capacity of 0 . 2 1 5 cal/g °C .

</think>

< answer >

A . 2 8 Kcal , 6 . 3 Kcal

</answer>

